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Open Grant Proposal: `Autonomous Interdependent Repositories - Basic Functionality Prototype`

Name of Project: Autonomous Interdependent Repositories

Proposal Category: `devtools-libraries`

Proposer: artem-v-shamsutdinov

Do you agree to open source all work you do on behalf of this RFP and dual-license under MIT and APACHE2 licenses?: Yes

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Project Description

- The problem:

Distributed Applications (DApps) lack an easy-to-use database layer.

The new DApps are gaining more functionality, getting more complex and currently the number of applications devs is limited due to a steep learning curve. For the majority of Software Engineers (who are used to working with relational databases) it is hard to write applications with complex data schemas.

- Our solution:

AIRport is a reference implementation of Autonomous Interdependent Repositories.

AIRport is a decentralized relational database of Repositories. Repositories are virtual databases, each with its own transaction log. Each Repository has a globally unique identifier that allows to distinguish it from other repositories in the same relational database (such as WebSql, or SqLite in a Cordova based App). For two Users to share a Repository it must be present on their devices, and the schemas used by that repository must be installed in AIRport databases on those devices.

```
For example, in the below diagram AIRport database contains 2
schemas,
one from App A and one from App B.  App A Schema contains table
A__TABLE_II
and App B Schema contains table B__TABLE_I.  3 repositories are
pictured.
Repository B1 spans both tables (and thus both schemas).  Repository
B2
is fully contained in B__TABLE_I.  Repository C1 spans only tables
in App
```

B Schema (other tables are not pictured).



AIR reposiories

Records in AIRport tables are always identified by 3 columns: * REPOSITORY_ID - this column contains the Repository's unique identifier * REPOSITORY_ACTOR_ID - The id of an "Actor" that created or modified the record * ACTOR_RECORD_ID - The id of the record (unique to combination of Repository and Actor)

Combination of these Ids allows for globally unique record identifiers within AIRport tables. It also allows to quickly find all records for a given repository or for a given Repository Actor. In practice, REPOSITORY_ACTOR_ID may also be unique while ACTOR_RECORD_ID will be unique to the device it was created on and may repeat in a given table because records may be created on different devices.

Each device/phone contains a single AIRport database that is shared by all applications on that device. The composition of the applications on each device can be different. The composition of the schemas installed in each AIRport database on each device can be different as well. Each database contains only the Repositories the user of that device decides to keep on it.



AIR across devices

Value

Our target users are developers for Distributed Applications. AIRport allows multiple applications to share the same schemas. It also allows to independently track data as Repositories. A Repository is a virtual database tracked in a number of tables across any number of schemas. At a high level, it's a "virtual interface" between applications and schemas.

This:

- Allows the Application End Users to seamlessly share the data only with selected Users.
- Lowers the barrier to entry for new Apps as they can reuse existing schemas or write add-on functionality to existing Apps.
- Lowers the barrier to entry by providing exiting data to new Apps.
- Allows hybrid applications where part of the data is centralized (for large scale sharing) and part is in private Repositories.

```
    For example, in an Event Tracking App, data for each ev
ent is a separate Repository.
    Other Applications can build upon this App's schema, pr
ovide additional schemas
    for functionality that builds upon the first App.  So,
another App can build
    a Event Chat System for participants and use both the E
vent Tracking App schema
    as well as it's own schema.  Yet another App can build
in Event Voting and use
    the schemas for both core Event Tracking and Event Chat
 and provide it's own
    schema as well.  Even more Apps can built add-on functi
onality without providing
    their own schemas.  The data used by all of these Apps
(across all related
    schemas) is stored in Event specific Repositories, one
repository per event.  Thus
    a repository spans multiple Event Apps and Schemas.  Th
e aggregate of specialized
    Event Apps together provides better functionality than
the sum of its part Apps.
    Therefore, AIRport enables synergies between Apps where
 "the whole is greater than
    the sum its parts", reducing the overall costs (and the
 costs of building each App).
```

- The benefits to getting this right

Each repository is completely autonomous and can be added to a host relational database and/or removed from it, at any time, without affecting other repositories. Repositories can have references to each other thus depending on data that other repository contains, but must be usable without referenced repositories (thus being both interdependent and autonomous).

AIRport Repositories have blockchain based transaction log storage to enable communication between devices.

AIRport offers refined, high productivity developer APIS:

- Simplified JPA annotations (no session concept, easier relations)
- GraphQL like query API
- GraphQL/Firebase like hybrid solution for mutation & access rules
- Automatic schema generation and installation

AIRport is fully functional off-line, commits are made locally and are added to the “longest chain” once device is back on-line.

- The risks:

Project is completely thought out and largely coded with a granted patent, risks of getting it wrong mostly relate to it being “developed in a vacuum”. In order to ensure that AIRport does not diverge from requirements of real applications we are developing a seed application - Votecube. Votecube has a complex relational schema and various persistence and validation needs.

The seed project is completely thought out and its most complex part is coded, the need for automatic reviews of articles based on multiple dimensions is clear and present (there are already working solutions out there with manual reviews from multiple points of view, we are now just taking it to the next level with automatic, and thus large scale, processing 3D interfaces, and aggregate analysis). The more basic break down interface will be based on the same (already thought out and implemented schemas) and will also be based on existing UI implementation (will add a simpler 3D axis based UI).

For example:

```
X-axis: Liberal - Conservative
Y-axis: Left Wing - Right Wing
Z-axis: Traditional - Progressive
```

The key risk is in figuring out how to monetize on the platform. To solve this we’ll be building a decentralized Advertisement Engine. Being decentralized it will allow full decentralized applications to be built on top of AIRport. Without it Apps will have to rely on centralized advertisement platforms for their revenue. It also makes long term success of the project possible, providing revenue to the AIRport team for bug fixes and improvements.

We have the experience to get the decentralized Advertisement Engine done:

- High definition model - Nik Dobrinov on
- AI and tensor models - Andrei Belitski

Economics

In order for AIRport to be successful all of the involved parties need to be incentivized to use it. The three primary party types are: - App Users - App Creators - AIRport Team

Users are generally interested in keeping their data private. They are also interested in reducing data duplication and duplicate data entry. Users may also be incentivized by profit sharing schemes from both App Creators and AIRport Team.

App creators are incentivized by the profits that applications generate. They are also incentivized in acquiring large market share (to maximize profits) by enhancing and enlarging their offerings but are dis-incentivized by the costs of additional work and expansion.

AIRport team incentives align closely with those application creators. They have an additional incentive to attract application creators to write Apps on the platform.

Airport naturally fulfils User data incentives though it’s technology. It also naturally solves the dilemma of large App monopolies by promoting App specialization (via schema+data sharing).

Monetization

A profit sharing setup will be used to reward schema creators when Apps use them. A nano-payment will be made the schema creator every time a CRUD operation is performed (by Apps that do not directly associate with that schema). Eventually the same logic will apply for BPMN logic providers.

To fulfil the rest of the incentives two alternative mechanisms will be employed:

Decentralized advertisement with profit sharing

Airport (in combination with Arweave) will support a privacy-centric advertisement engine. The Ad decision making engine will kept on user devices, thus solving data privacy issues. Data will not be shared with any authority or intermediary or the advertisers and will remain solely on the Users device (inside the relational database, which is embedded in the locally installed AIRport App). This keeps the User in full control of how much data is used for serving Ads.

The AIRport advertisement model is flipped from traditional push to pull. In the push model a central authority decides what Ads get pushed to a user. In AIRport pull model the User device decides what Ads to pull to the User. Ads themselves will be stored on Arweave and tagged to allow pulling the most appropriate Ads. Thus the advertisement platform becomes fully decentralized and cannot be manipulated by a central authority.

In the long term multiple incentive tiers will be employed for the Users, with increasing monetary rewards for all participants (because they provide increasingly accurate Ad placement):

1. Basic “content based” tier - users receive Ads based on the data that is (locally) sent to the Apps from AIRport (for any given rendered page).
2. Enhanced “content based” tier - Apps reinforces Ad choices by sending back to AIRport page metadata (and static content rendered on pages).
3. Demographic tier - user voluntarily stores their demographic data in AIRport. Again, it’s not shared with anyone, just used to pull more accurate Ads.
4. Geographic tier - users opt in to using their permanent and current location for serving Ads (again this information remains on the device only and is not shared with anyone).
5. Model tier - users opt in anonymized Big Data model generation for optimal Ad placement. This does require sharing data which is scrubbed of any PI (AIRport will provide annotations on per column basis to flag PI data).

Advertisement revenue will be split between the 3 parties (User, Apps, Airport) in a manner that best fits the market conditions at that time.

Fee model

Users can opt out (either wholesale, or per App, or per time period, or per repository) of advertisements by paying for usage of AIRport and applications. This allows the Apps and AIRport to still fulfill their monetary incentives. AIRport can act as either a central fee platform, where it shares User fee revenue with the Apps, or Apps can share their revenue with AIRport.

The expectation is that only a small number of Users will op-in for the fee model and that the bulk of funds will come from the decentralized advertisement engine.

Deliverables

- Artem Shamsutdinov

A relation database central to the device (which it is running on) with a localhost-only web API for native and web applications (as well as a web-only trial version of the application).

Shared virtual repositories (via independent transaction logs) based on IPFS DAG.

- Nikolay Dobrinov

A centralized (high precision) advertisement model based on anonymous user data.

- Andrei Belitski

A local tensor based (or newer tech if available) advertisement engine.

- Brian Gill

A seed application source: Full implementation of the seed project Votecube (Prototype)). AIRport documentation.

Development Roadmap

Phase 1: Basic Functionality Prototype

1. Month 1 - Core Framework

- Finish Non-Sharing core framework code (Observables disabled, basic schema generation, no indexes) - 3 weeks
- Debug core framework - 1 week

2. Month 2 - Basic Internal Functionality

- Finish basic repository creation & maintenance - 1 week
- Finish basic (semi-centralized) notification network - 2 weeks
- Debug and test all basic internal pieces to gether - 1 week

3. Month 3 - Conflict Resolutiona and test Apps

- Finish Basic Automated conflict resolution (timestamp based) - 1 week
- Write basic chat App - 1 week
- Write basic todo list App - 1 week
- Integration test applications with sharing - 1 weeks

4. Month 4 - Blockchain

- Write basic private blockchain implentation - 3 weeks
- Debug and integration test data blockchain protection - 1 week

5. Month 5 - IPFS integration

- IPFS integration with Javascript Client (for long term persistance) - 3 weeks
- Integration test with IPFS - 1 week

Functionality:

```
Basic Functionality Prototype
```

People:

```
1 person - Artem Shamsutdinov (framework developer)
```

Required organizational finding:

1. Month 1 - Core Framework

```
$10000 ($9000 Salary + Benefits, $1000 Operational Overhead)
```

2. Month 2 - Basic Internal Functionality

```
$10000 ($9000 Salary + Benefits, $1000 Operational Overhead)
```

3. Month 3 - Conflict Resolutiona and test Apps

```
$10000 ($9000 Salary + Benefits, $1000 Operational Overhead)
```

4. Month 4 - Blockchain

```
$10000 ($9000 Salary + Benefits, $1000 Operational Overhead)
```

5. Month 5 - IPFS integration

```
$10000 ($9000 Salary + Benefits, $1000 Operational Overhead)
```

Total: $50000

- Phase 1 timeframe

  - Project start date September 1st 2021
  - Project completion date February 1st 2022

  Total time: 5 months

- Phase 1 Budget

```
* Salary + Benefits
    Artem Shamsutdinov:                $45000
* Operational Overhead:                $5000

* Total cost:                          $50000
```

Maintenance and Upgrade Plans

This proposal builds a minimal functionality prototope. Subsequent proposals will be submitted for:

- Mobile device integration (Go + V8 + SqLite)
- Peer-to-peer notification network
- Peer-to-peer node registration network
- Repository joining and permissions maintenance
- Observable data (local to device)
- Fast notification mechanisms
- Schema versioning
- Advertisement engine
- (Global) repository search functionality

These will have to be completed in the following phases.

Team

Team Members

- Artem Shamsutdinov
- Nikolay Dobrinov
- Andrei Belitski
- Brian Gill

Team Member LinkedIn Profiles

- https://www.linkedin.com/in/artem-shamsutdinov-1b5910/
- https://www.linkedin.com/in/nikolaydobrinov/
- https://www.linkedin.com/in/dr-andrei-belitski-178b3a75/
- https://www.linkedin.com/in/brian-gill-1353b21/

Team Website

https://dataindependence.net

Relevant Experience

Dr. Nikolay Dobrinov - extensive experience in statistical model research and implementation

Dr. Andrei Belitski - extensive experience in AI, neural networks research and implementation

Brian Gill - overwhelming experience in application development

Artem Shamsutdinov - inventor of the technology and implementor of the core framework

Team code repositories

https://github.com/autonomous-interdependent-repositories

https://github.com/votecube

Additional Information

1. AIRport framework greatly aids development acceleration of distributed applications.

The process of installing AIRport is:

- User navigates to a consumer Application that uses AIRport and creates a Repository
- Application prompts the user to install AIRport database App (if not installed already)
- User installs AIRport App
- The consumer application creates the private Repository and prompts to add other participating users
- Creating user shares the Repository, and the App notifies new users



AIRport as Cordova application

AIRport Repositories have blockchain based transaction log storage to enable communication between devices. AIRport is fully functional off-line, commits are made locally and are added to the “longest chain” once device is back on-line. Each Repository transaction log is a separate chain and itself can consist of sub-chains if devices go out of sync. Thus, the Repository transaction log is a Directed Acyclic Graph with each commit being a separate block and all sub-chains are resolved to the “longest chain” via timestamp based conflict resolution mechanism.

```
For examle if Alice modifies record 1 while being offline and
Charlie
modifies the same record also offline but at a later time then both
with be notified of the conflict and automatic conflict resolution
will pick the latest column values while allowing for manual confict
resolution.
```



AIR Transaction Log

Based on my current limited understanding of blockchains:

A Repository chain can be integrated with IPFS since it supports DAG datastructure and is storage centric. Integrations with other chains are possible: A Repository chain can be tied to the chains of the individual chains of creators/participators in the Repository. In turn, those can be integrated into any 3.0 blockchain.

Core to AIRport is the idea is for multiple applications to collaborate and re-use data. The two key points here are:

- User is in control of their data they allow applications to access their data (usually only a part of their data).

- Applications are in control of sharing schemas with other applications
- they can allow or deny access to their schemas.

This the applications only interact with the device-local database, making AIRport fully operational in offline-mode. The database is in charge of maintaining the repositories contained in it. It may occasionally prompt the user to purge infrequently used repositories (or may do so automatically if not configured), leaving them only in the cloud backup. The applications however may request the user to load additional repositories from the cloud, for processing.

AIRport aims to provide zero code deployments where a thin shim is provided for the framework and TypeScript interfaces are used to define the query+mutation API against the schemas used by the application. The rest is defined in the schema configuration file, which is loaded into the device's AIRport database directly. The config file is generated from the annotated Entities and the DAO code, which itself relies on auto-generated TIQL (Typescript Instrumented Query Language) objects and interfaces as well as core framework code and generated Base DAO objects.

2. The time for Votecube decentralized decision-making system is at hand:

Existing businesses are doing manual reviews of articles from multiple points of view (centrist, left and right). We have the experience to do better with automated processing of a much larger number of articles and statistical analysis on the aggregate information. A new 3-Dimensional UI in the Votecube project will display 3 dimensional positional graph for each article and for all articles in on the topic. Dimensions could be many, but we'll be showing the ones most relevant for the topic (as determined by AI algorithms and per article automated statistical analysis). For example for political articles the dimensions could be:

X: Liberal - Conservative Y: Left - Right Z: Traditional - Progressive

Each individual article will be automatically evaluated for its position on these axes, and a graph will be presented for this article. An additional aggregate 3-D graph across all evaluated articles on the issue will present the overall picture of the public opinion.

The new UI will allow users to add their input on the article in terms of providing their own position of the article's position on the selected axis and allow adding comments on a forum.

Then based on distilled information from each article the "contrast cube" will be constructed. This aggregate cube will be based on the dimensions with the "strongest" ratings and will allow easy user voting on the topic.